

Repository onboarding, HPC deployment and self-healing

P04 Repository evidence inspection

SYSTEM	Act as a computational-biology MLOps engineer. Infer one candidate registry record from repository evidence for deterministic validation and human review.
USER	Given candidate evidence and repository context, return model name, environment hint, repository URL, Python version, dependencies, setup commands, inference template, weight evidence, confidence and unresolved risks. Leave unsupported commands empty.
VERIFIED OUTPUT	8 registry records currently persisted in data/local_registry.json
SOURCE	agents/model_onboarding/repository_inspector_system.md · repository_inspector_task.md

P05 Deterministic registry validation

SYSTEM	Treat Agent output as a proposal, not executable truth. Normalize dependencies, reject unsafe commands, preserve authoritative repository paths and require the declared FASTA/output placeholders.
USER	Validate required fields and evidence links before writing the registry. A model name alone is insufficient; repository, environment, inference command and weight status remain separately auditable.
VERIFIED OUTPUT	Registry write is separated from deployment readiness and formal benchmark admission
SOURCE	new_model_onboarding.py · model_resource_policy.py · data/local_registry.json

P06 Bounded HPC self-heal proposal

SYSTEM	Diagnose one failed environment or smoke test from registry, repository documentation and logs. Recommend the smallest evidence-supported repair. Deterministic code filters all packages, commands and registry updates.
USER	Return diagnosis, pip_install, conda_install, env_setup_commands, registry_updates, remove_requirement_patterns and retry_smoke. Never propose destructive or privileged commands.
VERIFIED OUTPUT	42 repair artifacts preserved under data/hpc_self_heal
SOURCE	agents/model_execution/self_heal_system.md · self_heal_task.md · hpc_model_ops.py

P07 Smoke-test admission gate

SYSTEM	Only an evidence-backed, environment-specific inference command may enter the smoke test. Failed imports, missing weights, missing databases and unsupported CLI arguments remain visible in the log.
USER	Run a mini-FASTA test, verify the output tree and prediction schema, retry only an approved bounded repair, and write readiness only after the smoke test succeeds.
VERIFIED OUTPUT	8 of 8 registry entries currently carry skip_env_setup=true
SOURCE	hpc_model_ops.py · data/hpc_self_heal/ · data/local_registry.json